Recap from last lecture

We want to calculate phase equilibrium in mixtures

To do this we need to be able to calculate the chemical potential of each species in each phase

You already know how to do this from mixture EOSs

But this is only possible for “simple” cases

We now need to learn how to do it for “difficult” cases

$$dG = -SdT + VdP \qquad (1)$$

$$\mu_i^L = \mu_i^V \qquad (5)$$

$$\mu_i = \underline{G}_i + RT \ln x_i \qquad (8,10)$$

One very simple example:

A liquid at equilibrium with its vapour in an inert, insoluble gas

What is the equilibrium vapour pressure?

Is it the same as for a pure liquid at equilibrium with its pure vapour?

e.g. water standing in a beaker in air at 20°C - what vapour pressure does it exert?

Is it the number I can look up in my steam tables?

From the Departmental Data Book (p52):

Properties of saturated water and steam:
temperatures from the triple point to 100°C

Temp. (°C)	Pressure (bar)	$\hat{V}$ (m^3/kg)		$\hat{U}$ (kJ/kg)		$\hat{H}$ (kJ/kg)	
		water	steam	water	steam	water	evap'n
0.01	**0.00611**	**0.001000**	**206.005**	**zero**	**2375.0**	**0.0**	**2500.9**
2	0.00706	0.001000	179.776	8.4	2377.6	8.4	2496.2
4	0.00814	0.001000	157.135	16.8	2380.4	16.8	2491.4
6	0.00935	0.001000	137.652	25.2	2383.1	25.2	2486.7
8	0.01073	0.001000	120.846	33.6	2385.9	33.6	2481.9
10	0.01228	0.001000	106.319	42.0	2388.6	42.0	2477.2
12	0.01403	0.001001	93.732	50.4	2391.4	50.4	2472.5
14	0.01599	0.001001	82.804	58.8	2394.1	58.8	2467.7
16	0.01819	0.001001	73.295	67.2	2396.9	67.2	2463.0
18	0.02065	0.001001	65.005	75.5	2399.6	75.5	2458.3
20	0.02339	0.001002	57.762	83.9	2402.3	83.9	2453.5
22	0.02645	0.001002	51.422	92.3	2405.0	92.3	2448.8
24	0.02986	0.001003	45.861	100.6	2407.8	100.6	2444.1
25	**0.03170**	**0.001003**	**43.340**	**104.8**	**2409.1**	**104.8**	**2441.7**
26	0.03364	0.001003	40.975	109.0	2410.5	109.0	2439.3
28	0.03783	0.001004	36.673	117.4	2413.2	117.4	2434.6

It is a good approximation that the EOS of water is unaffected by the presence of the air

However the mechanical pressure experienced by the liquid is no longer equal to the vapour pressure

- it is higher due to the presence of the air

We can draw exactly the same isotherm as for the pure case, but the equilibrium calculation has changed as follows:

Purple = no air present

Green = air present

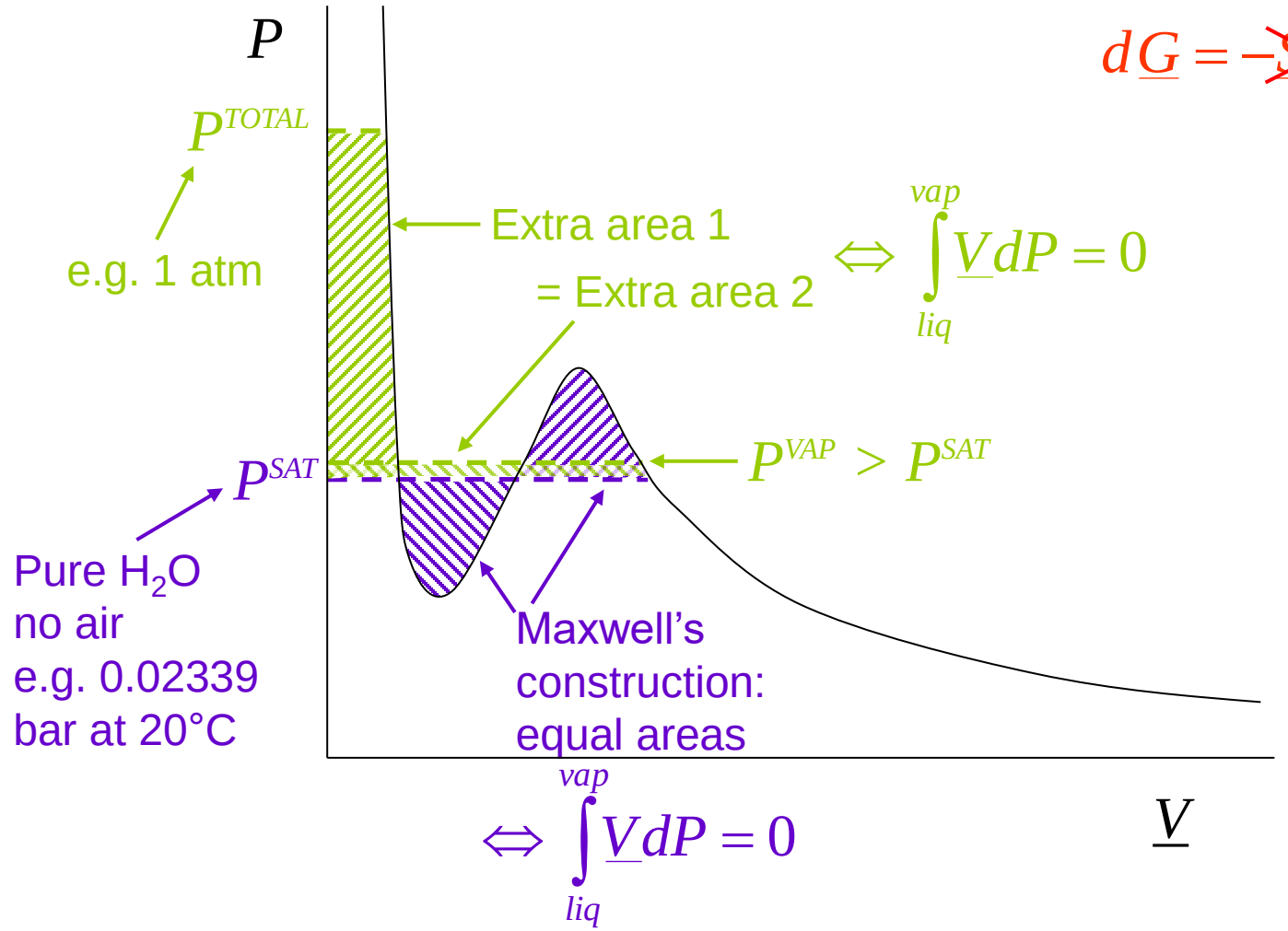
Thermal equilibrium – on isotherm

Mechanical equilibrium – same pressure

Chemical equilibrium $\underline{G}^v = \underline{G}^l$

$$d\underline{G} = -\cancel{\underline{S}dT} + \underline{V}dP$$

on isotherm



$T < T_c$ - same isotherm as pure H_2O (at same T)

At equilibrium in the presence of air we require,

Chemical equilibrium $\mu^l = \mu^v$

$\underline{G}^l = \mu^v$

Area (1) Area (2)

$$\int_{P^{SAT}}^{P^{TOTAL}} \underline{V}^l dP = \int_{P^{SAT}}^{P^{VAP}} \overline{V}_i{}^v dP$$

integrating along an isotherm

$\left(dG = -SdT + VdP\right)$

Liquids approx incompressible

P^{VAP} often low, so the vapour phase close to ideal gas.

$$\underline{V}^{l}\left(P - P^{SAT}\right) \approx \int_{P^{SAT}}^{P^{VAP}} \frac{RT}{P}\, dP = RT \ln \frac{P^{VAP}}{P^{SAT}}$$

$$P^{VAP} \approx P^{SAT} \exp\left[\frac{\left(P - P^{SAT}\right)\underline{V}^{l}}{RT}\right]$$

The Poynting correction

i.e. the vapour pressure exerted by a liquid is raised above the saturated vapour pressure of the pure liquid in the presence of other insoluble gases.

This is a small effect at ordinary pressures but can be significant at high gas pressures.

e.g. at 20ºC, P^{SAT} of water is 0.02339 bar (from steam tables) and

$$\underline{V}^{L} = 10^{-3}\ \mathrm{m^3 kg^{-1}} = 1.8 \times 10^{-5}\ \mathrm{m^3 mol^{-1}}$$

$$\frac{P^{VAP}}{P^{SAT}} \approx \exp(7.3\times10^{-4}) = 1.0007 \quad \text{at 1 atm (<0.1\% correction)}$$

$$\exp(0.0749) = 1.08 \quad \text{at 100 atm (8\% correction)}$$

Calculated the (partial) molar Gibbs energy of the vapour and liquid phases (of water) - they are equal at equilibrium

We ended up with an expression for the vapour pressure at equilibrium with the pure liquid phase

We assumed that the vapour phase was an ideal gas

The molar Gibbs energy of the liquid must be equal to the (partial) molar Gibbs energy of the ideal gas in equilibrium with it

This is a useful way of describing the molar Gibbs energy of non-ideal gases or liquids

We define a new concept; **fugacity, f_i**

The fugacity of a component (pure or in a mixture) is the partial pressure of an ideal gas of the same species which would be in equilibrium with that component





WE HEARD YOU THOUGHT THAT YOU UNDERSTOOD THERMODYNAMICS …

… SO WE INVENTED FUGACITY SO THAT YOUR PRESSURE WILL HAVE ANOTHER PRESSURE BUT IT'S REALLY NOT PRESSURE, SO WE'LL CALL IT FUGACITY

In the case of the Poynting correction:

Liquid fugacity, $f_i = P^{VAP} \approx P^{SAT} \exp\left[\frac{\left(P - P^{SAT}\right)\underline{V}^{l}}{RT}\right]$

'cos assumed IG

It is clear that fugacity has the units of [pressure]

Pure saturated liquid fugacity = P^{SAT} (assuming vapour is IG)

The relationship between fugacity and Gibbs energy

Gibbs energy is the important thermodynamic quantity

It is sometimes more convenient to use fugacity in calculations

Ideal Gas: $\underline{G}_i(T,P) = \underline{G}_i^0(T,P^0) + RT\ln\frac{P_i}{P^0}$

(partial) pressure of IG

pure IG at T, P^0

For a pure substance: $\underline{G}_i(T,P) = \underline{G}_i^0(T,P^0) + RT\ln\frac{f_i}{P^0}$

pure IG at T, P^0

Fugacity turns out to be particularly useful in mixtures

The fugacity of a component in a mixture is also called **fugacity,** but labelled $\overline{f}_i$

Each component present in a mixture has a fugacity.

This has exactly the same meaning as the fugacity for a pure substance: it is the partial pressure of an ideal gas which would be required to be in equilibrium with the component in the mixture. Fugacities may be defined for gases, liquids or solids.

For a component in a mixture, $\mu_i = \mu_i^0 + RT \ln \frac{\overline{f}_i}{P^0}$

↑ pure IG at T, P^0

and $\underline{G} = \sum \mu_i x_i$ $= \underline{G}_i^0$ because pure

Note that fugacity is not a <u>partial</u> <u>molar</u> <u>quantity</u>

i.e. $f^{TOT} \neq \sum \overline{f}_i x_i$ (f^{TOT} has no meaning)

{ but $\underline{G} = \sum \mu_i x_i = \sum \mu_i^0 x_i + RT \sum x_i \ln \frac{\overline{f}_i}{P^0}$ }

Don't worry about fugacity. It is a piece of convenient notation. There is no new thermodynamics in fugacity - it just takes some getting used to.

The important concepts to understand are Gibbs energy and chemical potential.